

Jordan Firth

Lieutenant Colonel, USSF

jordan.firth@afacademy.af.edu

704-219-8918

EDUCATION

North Carolina State University

Raleigh, NC

PhD, Aerospace Engineering

2016 Jul–2019 Aug

GPA: 3.89

Dissertation: Deployment Mechanisms for Rollable Spacecraft Booms

Performed Zero-G parabolic test flights of two novel deployment mechanisms for rollable spacecraft booms with funding from a NASA flight opportunity grant. Led a 14-member team to design, develop, and manufacture instrumentation, support equipment, and booms and deployment mechanisms, including satisfaction of FAA airworthiness standards.

US Air Force Institute of Technology

Wright-Patterson AFB, OH

MS, Astronautical Engineering

2009 Aug–2011 Apr

GPA: 3.68

Thesis: Vibration Interaction in a Multiple Flywheel System

Focus areas: astrodynamics and spacecraft system design. I was the team lead for a two-semester student project building and testing a cubesat-sized academic spacecraft. For my thesis I investigated the effects of flywheel vibration in spacecraft structures.

US Air Force Academy

USAF Academy, CO

BS, Engineering Mechanics

2002 Jun–2006 May

GPA: 3.36

Focus areas: composites and aircraft structures. For my senior design project I designed a wing-folding solution for smaller packaging of an operational AFSOC micro air vehicle.

UNDERGRADUATE EDUCATOR EXPERIENCE

Department of Astronautics

USAF Academy, CO

Instructor, Course Director, Astro 331: Space Systems Engineering 2022 Aug–Present

Directed intermediate lab-based space systems engineering course. Students gain fluency in all major spacecraft subsystems and system-level design issues. Transitioned lab curriculum to a sustainable experience based on commodity hardware. Cadets build a breadboard-based spacecraft analog and perform lab experiments to reinforce their learning. Required for all 3rd-year Astronautical Engineering Majors.

Department of Astronautics

USAF Academy, CO

Instructor, Course Director, Astro 321: Intermed. Astrodynamics 3 sem, 2015–2023

Instructed intermediate course in orbit mechanics. Students gain familiarity with orbital mechanics including orbit determination, maneuvers, perturbations, and rendezvous and proximity operations, as well as programming skills to perform these operations at scale. Required for all 3rd-year Astronautical Engineering Majors.

Department of Astronautics

USAF Academy, CO

Instructor, Course Director, Astro 310: Intro to Astro 6 sem, 2014–Present

Directed introductory astrodynamics, including orbital mechanics, spacecraft systems, and rocket propulsion. Required for all Air Force Academy cadets.

AFROTC Detachment 595

NC State University, Raleigh, NC

Course Director, AS 222: Team and Leadership Fundamentals II 2018 Jan–2018 Apr

Taught spring semester sophomore-level ROTC leadership class to 27 students/cadets at the request of the Det commander while completing a PhD at NCSU. Topics include AF history and personal and team leadership skills. Course prepares students/cadets to put leadership concepts into practice at field training. Required for all AF ROTC cadets.

Department of Engineering Mechanics

USAF Academy, CO

Instructor, EM 320: Dynamics 2016 Jan–2016 May

Instructed introductory dynamics, covering kinematic and kinetic analysis of particles and rigid bodies, as well as an introduction to mechanical vibrations of simple systems. Topics include kinematics using Cartesian, path, and polar coordinates; kinetics using force-mass-acceleration, work-energy, and impulse-momentum methods. Required for all Aeronautical, Astronautical, and Mechanical Engineering Majors.

PROFESSIONAL EXPERIENCE

Department of Astronautics

USAF Academy, CO

Assistant Professor

2022 Aug–Present

Instructor for 3 courses. Course director for Astro 331: Space Systems Engineering; assumed responsibility for ongoing multi-year overhaul. Rewrote Arduino-based lab curriculum with a renewed emphasis on student ownership of lab exercises.

Developed a 40-hour spacecraft systems short course based on Astro 331. Coordinated a Cooperative Research and Development Agreement (CRADA) with Singapore's Space Faculty, Ltd., including approval from SAF/IA and SAF/GC. Delivered course to 9 STEM students, with plans for future follow-on collaboration.

Mentor to multiple FalconSAT capstone student research groups, including flywheel energy storage, coaxial cable bending, and reaction wheel manufacturing. Led reaction wheel team through balancing and vibration test campaign. Identified and corrected design flaws in multiple tests.

Director of Assessments

2022 Aug–Present

Responsible for maintaining department assessment program supporting continuous improvement and periodic ABET accreditation. Supported 2022 Engineering Program Advisory Council. Performed mid-cycle ABET self-assessment; identified and filled minor gaps in department assessment program.

DIA Defense Technology and Long-Range Analysis Office

Charlottesville, VA

Chief, Acquisition Intelligence Branch

2020 Dec–2022 Jul

Branch chief for 29 acquisition intelligence analysts at seven sites, including contractors, DIA civilians, reservists, and joint active duty officers from 4 services. My branch assessed near-term threats posed to US and allied systems by emerging adversary technology. We also maintained the authoritative defense intelligence threat library, which connected program staff for \$1.23T worth of defense programs to intelligence authors, ensuring that US capabilities would not become obsolete before fielding.

As the military advisor for a 150-person office, I also guided development of eleven assigned officers, matching individual skills to organizational needs to ensure effective use of each military officer during their three-year DIA assignment.

Air and Space Analysis Team Lead

2019 Aug–2020 Nov

Coordinated with program offices and requirement owners threat assessments affecting acquisition programs. Co-wrote 20-year analytic outlook of space-based threats to US and allied systems.

Department of Astronautics

USAF Academy, CO

Instructor, Engineering Division Executive Officer

2014 Jul–2016 Jul

I taught engineering classes in two academic departments (Astronautics and Engineering Mechanics) over five semesters, including an astronautics class required for all students. I developed curriculum for three classes as course director. I also trained incoming faculty during summer academics.

Infrared Space Systems Directorate

Los Angeles AFB, CA

SBIRS System Test Engineer

2012 Mar –2014 May

I oversaw system test for the SBIRS missile warning system, including pre- and post-launch checkouts for three spacecraft. This testing led to successful completion of a major in-place upgrade to the SBIRS ground system with no downtime.

I also compiled and edited the biennial update to the Congressionally-mandated system Test and Evaluation Master Plan and coordinated it with external agencies for approval by the Office of the Secretary of Defense.

DSP Orbital Operations Officer

2011 Apr–2012 Mar

I oversaw execution of the ongoing factory support contract which provided vital manufacturer expertise for operators of the DSP missile warning satellite constellation. This support was necessary to provide continuous global missile warning. Contractor activities included training all incoming personnel, anomaly resolution, and spacecraft health monitoring.

During this assignment we negotiated a new support contract, ensuring continued coverage while planning for a graceful transition as DSP was replaced with the next generation of missile warning spacecraft. I also provided senior directorate leaders with regular updates on the health of the constellation.

846 Test Squadron

Holloman AFB, NM

Rocket Sled Test Engineer

2008 Sep–2009 Aug

Performed Engineering analysis of rocket sled test profiles at Holloman High Speed Test Track, the nation's premier rocket sled track. Responsible for technical test details including sled structural margins, acceleration profiles, and rocket staging. This work required coordination with program managers, in-house fabrication shops, track operators, and support personnel.

During this assignment I revived a dormant squadron capability to launch captive Stinger missiles down the track, providing accurate threat profiles for in-flight testing of infrared detectors for inbound missiles. We used this capability to provide the final operational checkout of a new CH-47 countermeasure system before its fleet-wide deployment.

586 Flight Test Squadron

Holloman AFB, NM

Flight Test Engineer

2006 Aug–2008 Sep

Performed advanced avionics flight test and test support using highly-modified T-38B and C-12J aircraft. These aircraft were used to both carry test equipment and serve as targets for urgent DOD priority missions.

I planned photo/safety chase for the final pre-deployment test of JASSM-ER, which verified the weapon's safe separation from the launching aircraft and a successful 500+ mile flight. In addition, I coordinated the final pre-delivery test of Iraq's first air-to-ground missile capability—Hellfire missiles launched from a Cessna Caravan.

RESEARCH EXPERIENCE

FalconSAT reaction wheel

2022 Aug–Present

Led cadet teams testing a cadet-built spacecraft reaction wheel. Balanced wheel on a custom-designed balancing fixture, and upgraded fixture to increase measurement accuracy. Performed qualification-level vibration test to determine launch survivability. Redesigned reaction wheel housing to eliminate design flaws revealed during vibration testing.

FlatSAT educational spacecraft

2022 Aug–Present

Developed a low-cost educational spacecraft analog for teaching systems engineering. This Arduino-based platform currently includes XBee radio communication, magnetometers and reaction wheel attitude control, solar panels and electrical characterization, and a visible light camera. Cadet collaborators have added sun sensors for attitude determination and on-board data storage. FlatSAT was developed for a semester-long undergraduate space systems engineering course, but has also been used for a week-long course tailored for STEM-interested undergraduates and educators.

Zero-G parabolic flight test

2017 Aug–2019 Aug

Developed 2 deployment mechanisms for Rollable spacecraft booms. Led a 14-member team to successful completion of two Zero-G parabolic test flights. Coordinated all aspects of flight testing, including structural, electrical, and chemical safety; instrumentation and analysis; researcher training; and fabrication and assembly of commercial, machined, and composite components.

3-DOF Air Bearing Attitude Platform

2023 Jun–Present

Hardware implementation advisor for four cadets performing independent research on an air bearing platform for spacecraft attitude research. Ensured personnel safety and effective resource allocation.

PUBLICATIONS

Burkhart, Devin M., John R. Snider, and **Jordan A. Firth**. "Flight Qualification for Low Cost Student Manufactured and Balanced Reaction Wheel." *Unpublished, presented at 2023 AIAA Region V student conference*. (2023)

White, Charles, **Jordan Firth**, and Mark Pankow. "Impact of Joint Parameters on Performance of Self-Opening Dual-Matrix Composites." *Journal of Spacecraft and Rockets* (2023): 1-12.

Adamcik, Benjamin, et al. "Impact of storage time and operational temperature on deployable composite booms." *AIAA Scitech 2020 Forum*. 2020.

Firth, Jordan A., and Mark R. Pankow. "Minimal unpowered strain-energy deployment mechanism for rollable spacecraft booms: ground test." *Journal of Spacecraft and Rockets* 57.2 (2020): 346-353.

Firth, Jordan A., and Mark R. Pankow. "Advanced dual-pull mechanism for deployable spacecraft booms." *Journal of Spacecraft and Rockets* 56.2 (2019): 569-576.

Firth, Jordan, et al. "Minimal Unpowered Strain-Energy Deployment Mechanism for Rollable Spacecraft Booms." *AIAA Scitech 2019 Forum*. 2019.

Brown, Robert B., and **Jordan A. Firth**. "Analysis of trans-Neptunian objects and a proposed theory to explain their origin." *Monthly Notices of the Royal Astronomical Society* 456.2 (2016): 1587-1594

Firth, Jordan, and Jonathan Black. "Vibration interaction in a multiple flywheel system." *Journal of Sound and Vibration* 331.7 (2012): 1701-1714.

AFFILIATIONS

American Institute of Aeronautics and Astronautics (AIAA) - Member

CERTIFICATIONS

DAWIA Test & Evaluation Level 3

Fundamental of Engineering Exam completion

COMMUNITY

USAFA/COS Baptist Student Union Alumni Association

Volunteer 3 years, 2014–Present

Led and participated in weekly Bible study for 10–30 cadets, and occasionally spoke to a larger group of 60+ cadets.

Vice President 2014 Aug–2016 Jun

Oversaw \$120,000 annual ministry budget as we completed filing as a tax-exempt organization. Provided board oversight for 3 staff ministry leaders. Built relations between alumni association and current students as the ministry grew to four local college campuses.

Grace Bible Camp Goshen, VA

Volunteer 2020–2021

Led a major renovation project involving 24 volunteers over two weeks to repair multiple facilities. Added egress routes, fixed HVAC systems, and completed structural repairs. Worked with camp staff and board members to improve facilities used by 350 campers/year.

Fairview Baptist Church Apex, NC

Nominating Committee Member 2018–2019

Selected church member for open positions. Helped senior pastor develop a plan for church growth, including a communication plan to ensure community buy-in.

Adult Sunday School Leader 2017–2019

Served as Sunday School teacher for a group of 20 adult members and their children. Taught weekly lessons and built relationships between group members.

YMCA Youth Soccer Apex, NC

Coach 5 seasons, 2016–2019

I was a soccer coach for six teams over five seasons. Players ranged in age from 4–9 and included many first-time players. I led them in weekly practices and games, ensuring they had fun and learned how to work together on a team.

REFEREES

Lt Col Luke Hagen

Department Head, USAFA Department of Astronautics, luke.hagen@afacademy.af.edu, 719-333-4110
Supervisor, 2023

Col Luke Sauter

Permanent Professor, USAFA Department of Astronautics, luke.sauter@spaceforce.mil, 719-671-9781
Supervisor, 2022–2023

Bill O'Hara

Chief, DIA Acquisition Intelligence Division, william.o'hara@dodiis.mil, 434-956-2166
Supervisor, 2020–2022

Tim Smith

Senior Intel Officer, DIA Acquisition Intelligence Division, timothy.smith2@dodiis.mil, 434-956-2408
Analytic Mentor, 2019–2022

Mark Pankow

Assoc. Professor, NCSU Mechanical & Aerospace Engineering Dept, mrpankow@ncsu.edu, 510-384-6064
PhD Advisor, 2016–2019